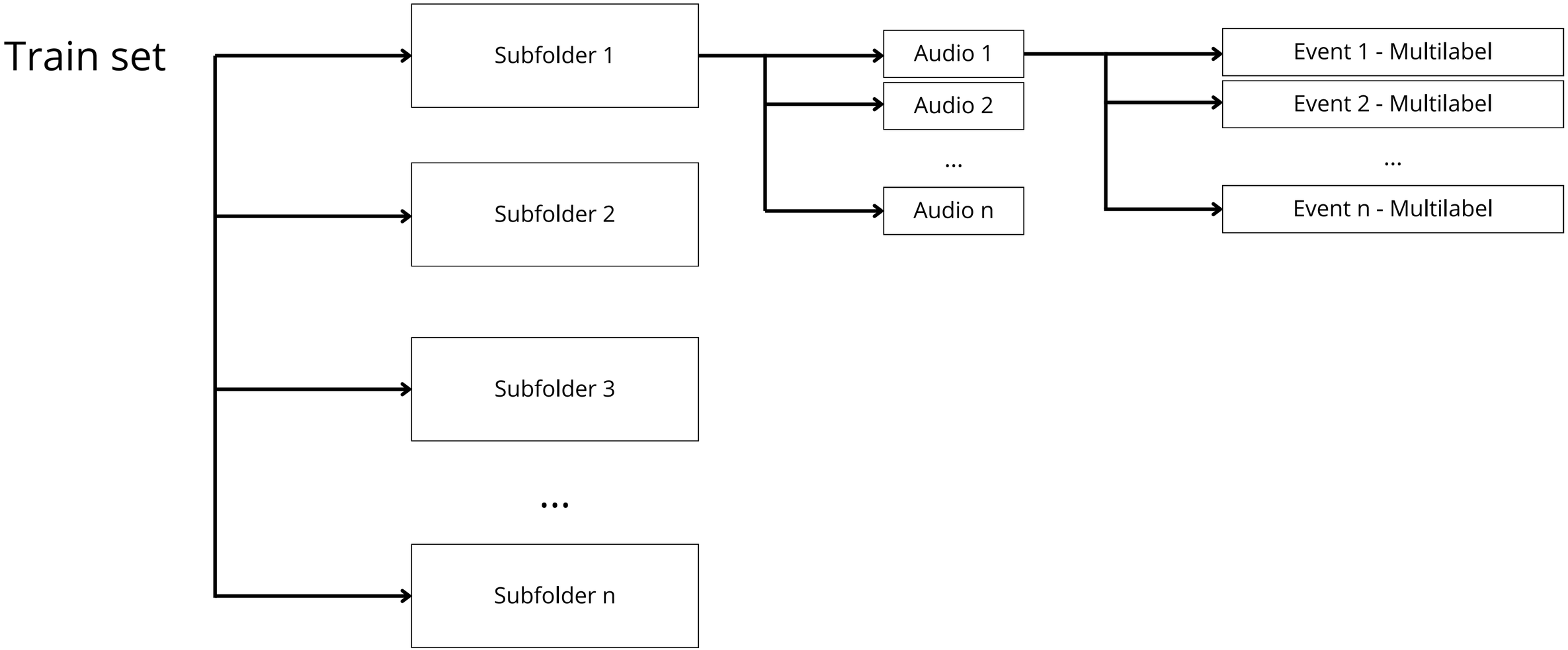
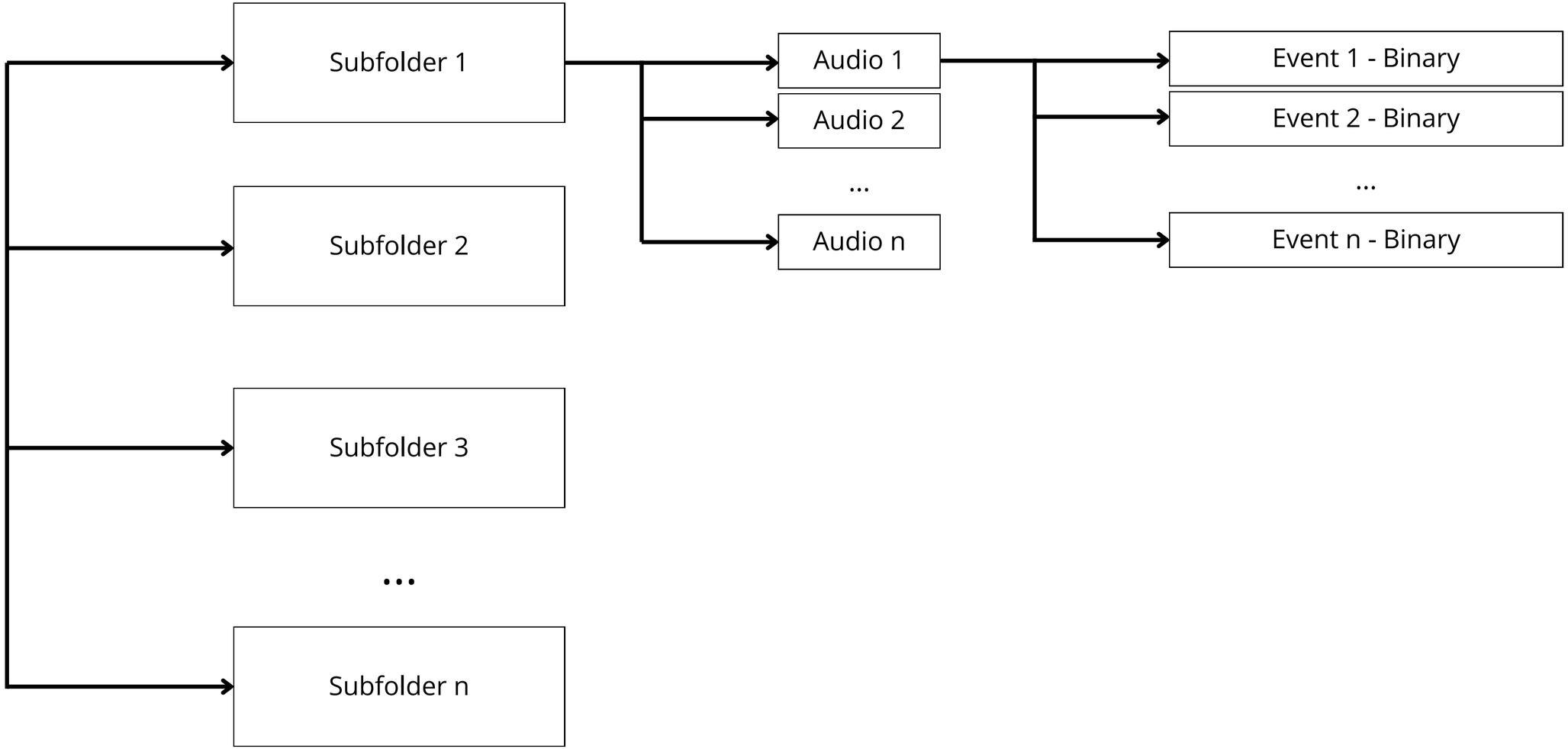
Train set

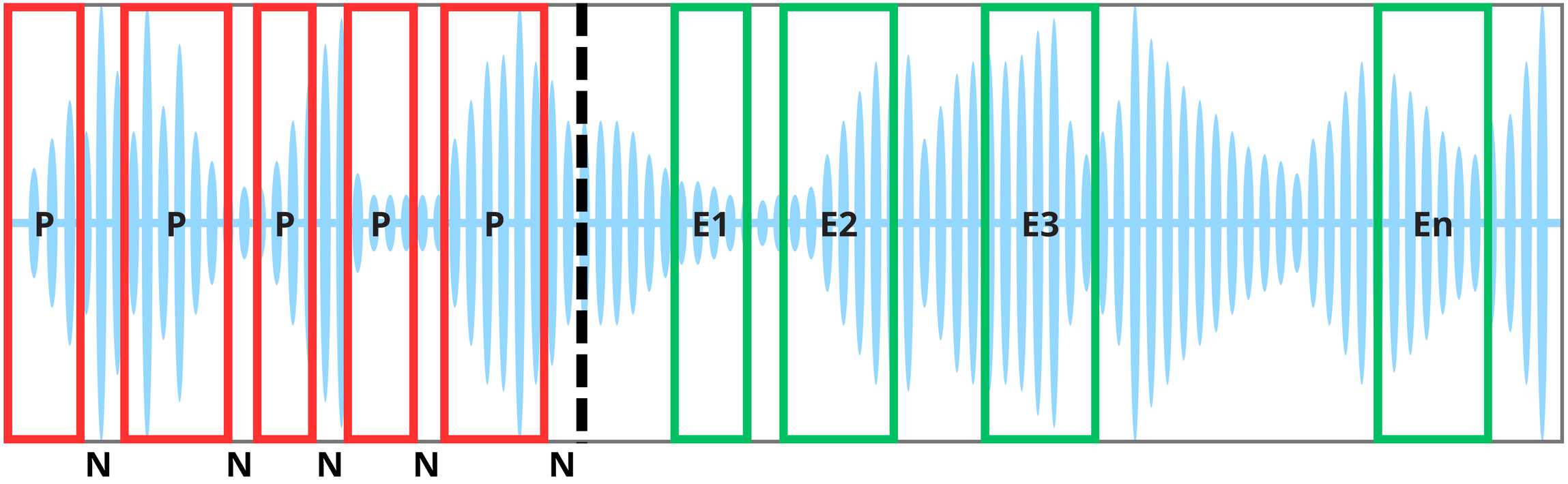

Train set
Subfolder 1
Subfolder 2
Subfolder 3
...
Subfolder n
Audio 1
Audio 2
...
Audio n
Event 1 - Multilabel
Event 2 - Multilabel
...
Event n - Multilabel

Val set

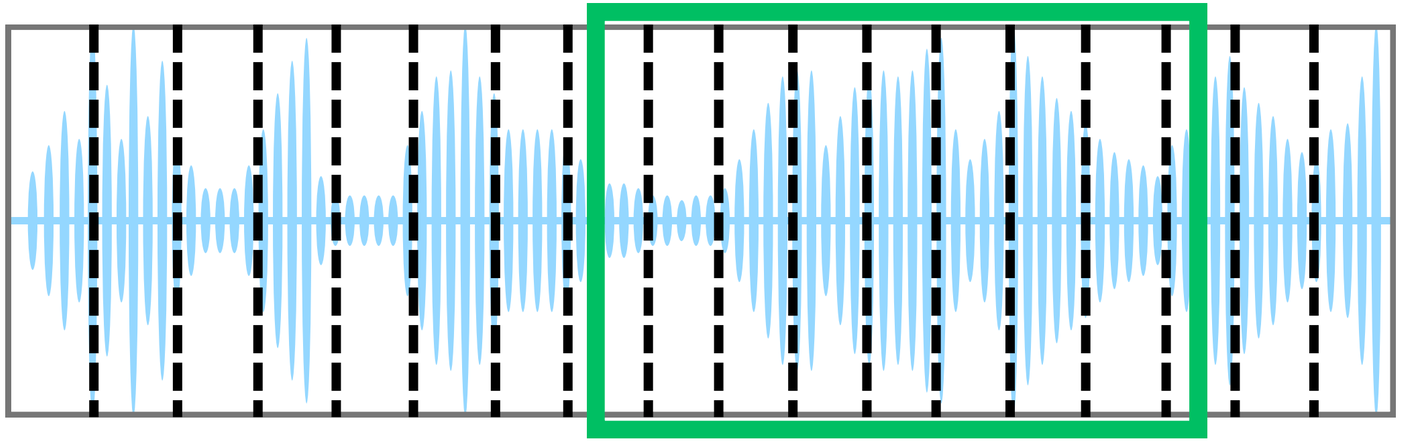
Val set



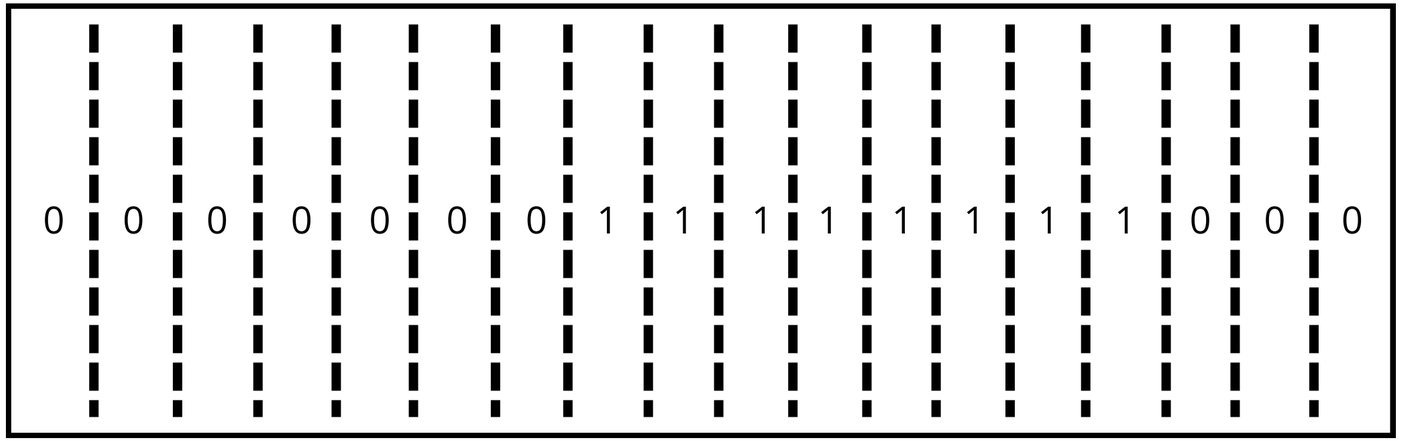
Evaluation



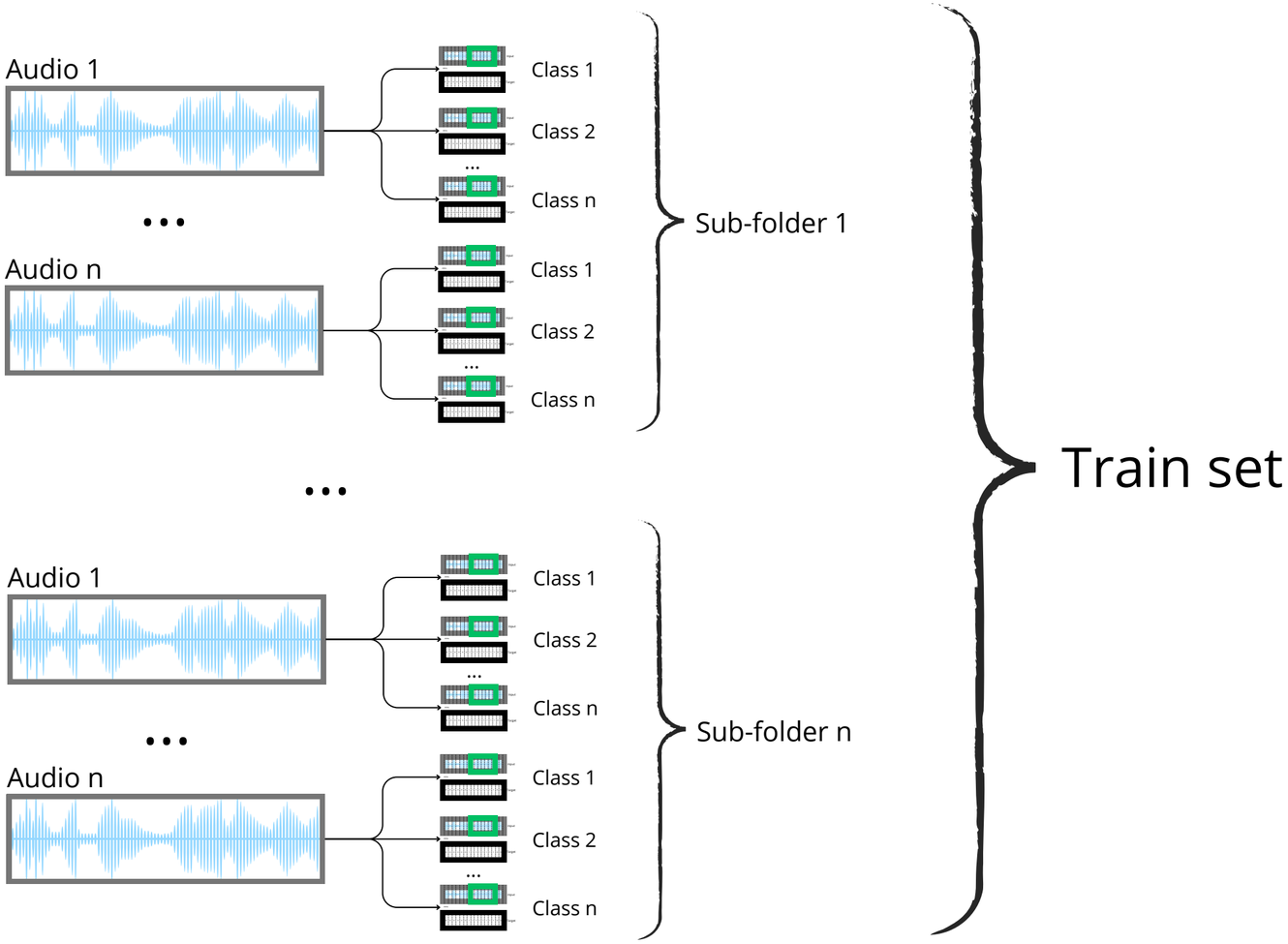
Possible approach



Input length = Target length

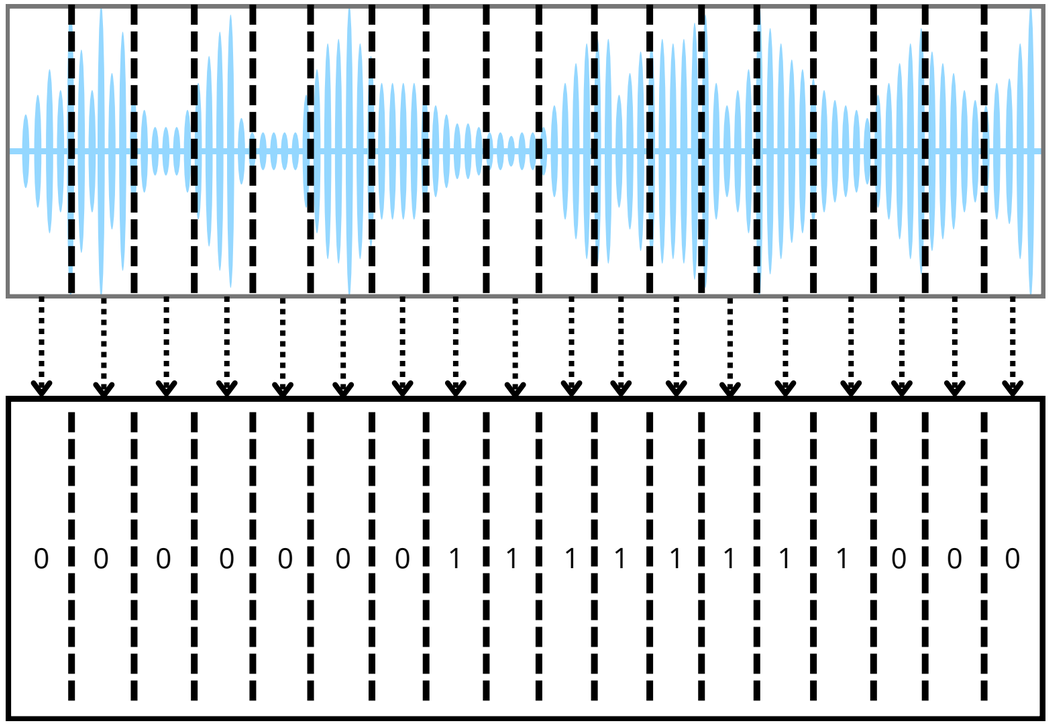


Multitask approach



- Each class is a task
- Multi-task: approach having a number of tasks equal to the total number of classes in the training set
- If an audio contains more than one class it will be replicated for a number of times equal to the total number of classes within it.

Training approach: 1st objective



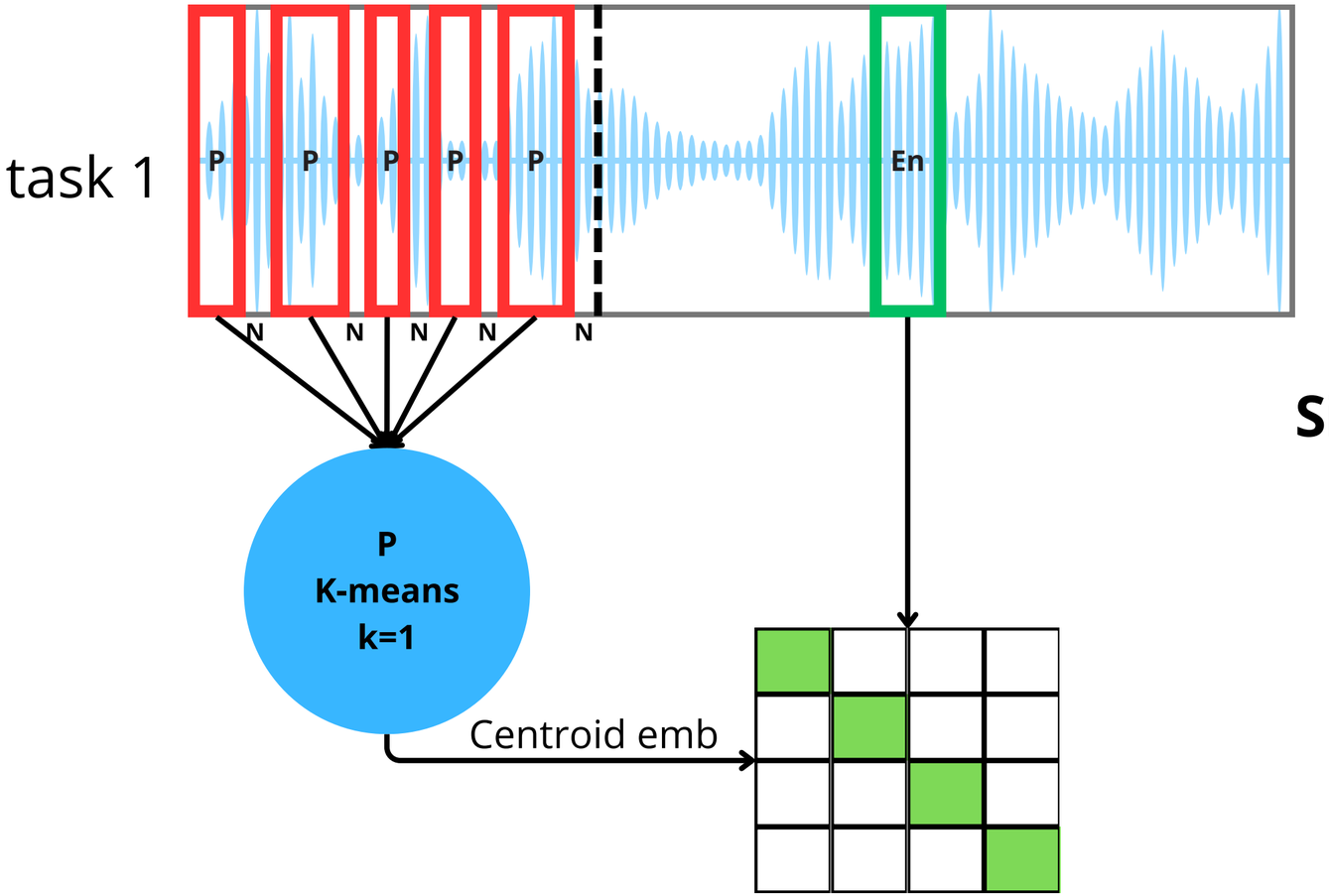
Input

task 1

Binary Classification: CE1

Target

Training approach: 2nd objective



Similarity Maximization: CE2

Training approach: Task objective

$$L_1 = CE_{11} + CE_{12}$$

$$L_2 = CE_{21} + CE_{22}$$

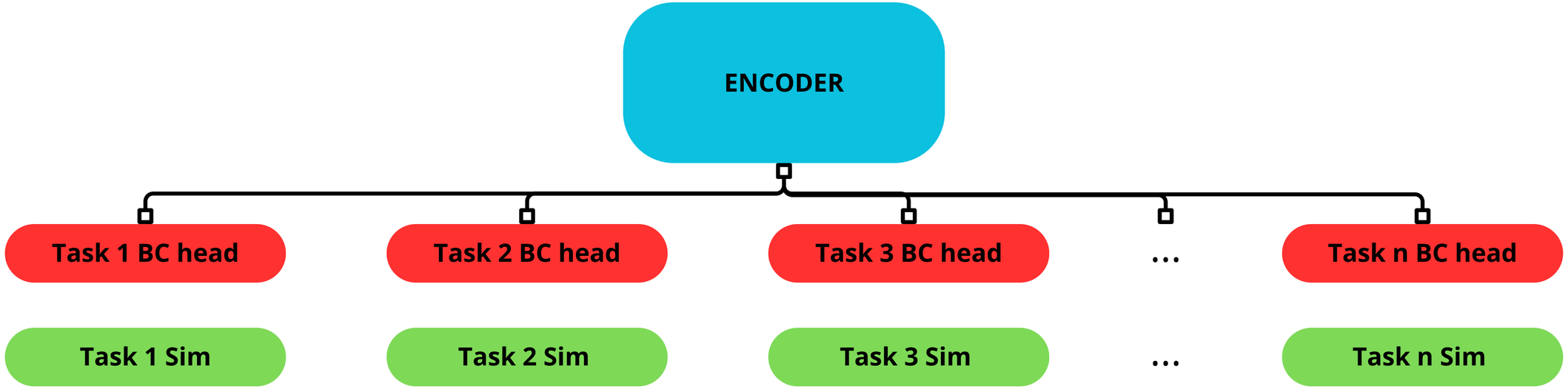
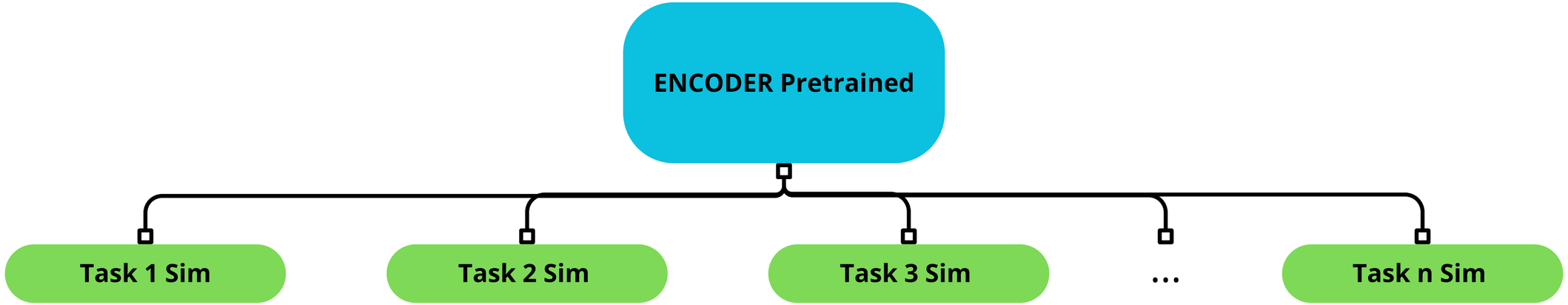
$$\cdots$$

$$L_n = CE_{n1} + CE_{n2}$$

Training approach: Total objective

$$L = \sum_{i=1}^{n} (L_i)$$

Model architecture on training set



Model architecture on validation set